

CO543/CO5430 Computer Vision Project Proposal - 2026

Project ID: P17 | Category: Industrial Inspection | Group: G08

Title: Automated Industrial Surface Defect Detection and Localization Using Deep Anomaly Detection

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1. Problem Statement and Motivation

Surface defect detection is a critical quality control task in manufacturing. Manual inspection is slow, subjective and error-prone under high-throughput conditions. Automated detection using computer vision can reduce waste, lower costs and improve consistency. However, defective samples are rare in practice, making supervised approaches impractical. Unsupervised anomaly detection addresses this by training only on defect-free images and flagging deviations at test time. This project compares a reconstruction-based baseline (Convolutional Autoencoder) with a feature-embedding method (PatchCore) on the MVTec AD benchmark, covering both detection and pixel-level localization. An explainability layer (Grad-CAM and occlusion sensitivity) will provide interpretable defect attribution.

2. Related Work

Bergmann et al. (2021) introduced the MVTec AD dataset with pixel-level masks for 15 industrial categories, establishing the standard anomaly detection benchmark. Reconstruction-based methods such as autoencoders detect anomalies via reconstruction error but struggle on textures (Bergmann et al., 2019). PaDiM (Defard et al., 2021) models pretrained patch features with Gaussian distributions. PatchCore (Roth et al., 2022) uses a coreset-reduced memory bank of pretrained features and achieves near-perfect image AUROC on multiple categories. This project compares reconstruction-based and feature-based approaches and adds quantitative explainability using Grad-CAM (Selvaraju et al., 2017) and occlusion sensitivity (Zeiler and Fergus, 2014).

3. Dataset

The MVTec Anomaly Detection dataset (CC BY-NC-SA 4.0) contains 5,000+ high-resolution images across 15 categories. Each category has defect-free training images and test images with pixel-level ground truth masks. Three categories are selected: Bottle and Hazelnut (objects) and Carpet (texture), covering structurally different defect types.

Item	Description
Source / License	mvtec.com/company/research/datasets/mvtec-ad CC BY-NC-SA 4.0
Selected Categories	Bottle (object), Hazelnut (object), Carpet (texture)
Training / Test	Defect-free images only (train) Normal + defective with pixel masks (test)
Preprocessing	Resize 256x256, normalize to ImageNet mean/std

4. Proposed Method

Baseline - Convolutional Autoencoder: Trained on defect-free images to learn normal appearance. Anomalies produce higher reconstruction error (MSE). The per-pixel error is the anomaly map and the image-level maximum error is the anomaly score. **Improved Method - PatchCore (WideResNet50):** Extracts patch-level features from a pretrained ImageNet backbone into a coreset-reduced memory bank. Test patches are scored by nearest-neighbour distance to the bank. Requires no gradient-based training on the target data. **Explainability:** Grad-CAM for the autoencoder (gradients accessible) and occlusion sensitivity for PatchCore (anomalib disables gradients internally). Both are quantitatively scored against ground truth masks.

Aspect	Baseline (Autoencoder)	Improved (PatchCore)
Features	Learned from scratch	Pretrained ImageNet (WideResNet50)
Anomaly Signal	Reconstruction error (MSE)	Nearest-neighbour distance in feature space
Training	Gradient-based (~100 epochs)	Memory bank construction (no training)
Explainability	Grad-CAM	Occlusion sensitivity

5. Evaluation Plan

Metric	Level	Purpose
Image AUROC	Image	Binary classification: normal vs. defective
Pixel AUROC	Pixel	Per-pixel localization quality
Per-Region Overlap (PRO)	Pixel	Size-balanced localization evaluation
Pointing Game Accuracy	XAI	Highest-attribution pixel inside defect region
Top-5% IoU	XAI	Overlap of top 5% attribution pixels with ground truth

An ablation comparing ResNet18 vs. WideResNet50 backbones will analyze the localization-vs-speed trade-off. Qualitative outputs (heatmaps, predicted masks, failure cases) will also be produced.

6. Project Goals

Level	Description
Minimum	Autoencoder baseline on Bottle. Report Image AUROC and Pixel AUROC with basic heatmaps.
Expected	Both methods on all 3 categories. All 5 metrics. Backbone ablation. Explainability analysis.

Level	Description
Stretch	Additional MVTec AD categories. Real-time inference demo. Lightweight deployment model.

7. Timeline

Week	Period	Tasks	Milestone
1	1-7 Jul	Group formation, topic registration, GitHub setup	M0
2	8-14 Jul	Finalize proposal, begin dataset download and EDA	M1
3-4	15-28 Jul	Preprocess MVTec AD, implement and evaluate autoencoder baseline	M2
5-6	29 Jul-11 Aug	Implement PatchCore, run on all categories, compute metrics	-
7	12-18 Aug	Ablation study, implement Grad-CAM and occlusion sensitivity	M3
8	19-25 Aug	Complete experiments, generate qualitative outputs, finalize tables	M4
9	26 Aug-1 Sep	Write IEEE report, clean repository, prepare demo	M5
10	2-7 Sep	Final submission, presentation slides, demonstration	M6

8. Risk Assessment

Risk	Impact	Mitigation
GPU memory insufficient for WideResNet50	Medium	Fall back to ResNet18; use CPU with longer runtime
Autoencoder fails on texture categories	High	Expected per literature; document as finding and analyze why
PatchCore memory bank exceeds RAM	Medium	Reduce coreset percentage (built into anomalib)
Explainability produces poor localization	Medium	Analyze failure modes; compare multiple occlusion patch sizes
Team member unavailability	Low	Clear task division; shared repo; weekly sync

References

- [1] P. Bergmann et al., "The MVTec Anomaly Detection Dataset," IJCV, vol. 129, pp. 1038-1059, 2021.
- [2] K. Roth et al., "Towards Total Recall in Industrial Anomaly Detection," CVPR, 2022.
- [3] T. Defard et al., "PaDiM: A Patch Distribution Modeling Framework for Anomaly Detection," ICPR, 2021.
- [4] R. R. Selvaraju et al., "Grad-CAM: Visual Explanations from Deep Networks," ICCV, 2017.
- [5] M. D. Zeiler and R. Fergus, "Visualizing and Understanding Convolutional Networks," ECCV, 2014.
- [6] P. Bergmann et al., "Improving Unsupervised Defect Segmentation by Applying Structural Similarity to Autoencoders," VISIGRAPP, 2019.